Reactions Between Acids and Bases

Neutralization, titration, and staying in balance

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Standards

Chemistry 2.6 — Acid–Base Chemistry: neutralization, titration, hydrolysis, and buffers.

Objectives

1. What happens when an acid and a base react?
2. How does titration determine an unknown concentration?
3. How do buffers resist pH change?

Expectations

1. Write and balance neutralization equations
2. Interpret a titration curve
3. Explain how a buffer works

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What Do You Already Know?

WHAT I KNOW	WHAT I WANT TO KNOW	EXPLORE THE SOURCE
▪ Antacids neutralize stomach acid ▪ Vinegar + baking soda fizzes and calms ▪ Blood pH barely changes even after a workout	▪ Why does mixing acid + base feel “safe”? ▪ How do chemists find an unknown concentration? ▪ What keeps blood pH so stable?	► PhET: Acid-Base Solutions ► PhET: pH Scale ► Khan Academy: Acids & Bases *Tap a link to explore before class discussion.*

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Neutralization Reactions

- A neutralization reaction: acid + base → salt + water.
- $HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$
- A salt is an ionic compound made of a cation from the base and an anion from the acid.
- It's a double-replacement reaction — H^+ and OH^- "cancel out" by forming water.
- Analogy: two dance partners swap — the acid's H^+ pairs with the base's OH^- to become water, while the leftover ions pair up as the salt.

$$H^+ + Cl^- + Na^+ + OH^- \rightarrow$$

$$Na^+ + Cl^- + H_2O$$

Salt = NaCl

Water = H_2O

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Titration: Finding an Unknown Concentration

1. Place a measured volume of unknown acid/base in a beaker; record initial pH.
2. Fill a burette with the titrant — a solution of known concentration.
3. Add titrant slowly, mixing and recording pH after each addition.
4. Continue until the equivalence point: moles H^+ = moles OH^-.
5. Use Ma × Va = Mb × Vb to solve for the unknown molarity.

$$Ma \times Va = Mb \times Vb$$

Key Terms

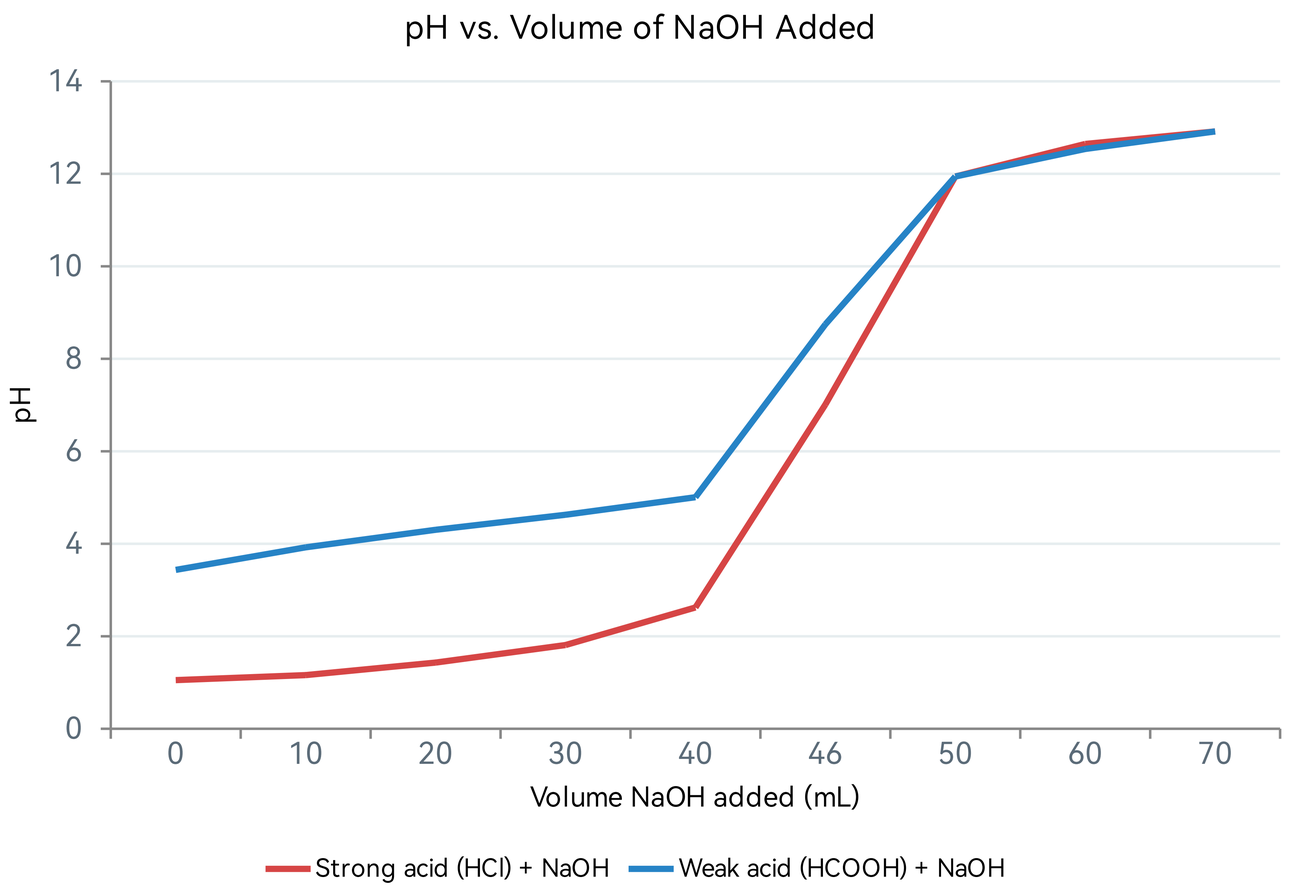
Equivalence point: moles H^+ = moles OH^-

End point: when the indicator changes color

Indicator: a dye that changes color across a pH range

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Titration Curves: Strong vs. Weak Acid



Reading the curves

- Both curves jump steeply at the equivalence point (~46 mL here).
- The strong-acid curve starts lower and jumps more sharply.
- The weak-acid curve has a gentler buffer region before the jump.

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Acid–Base Indicators

Chemical dyes whose color depends on the pH of the surrounding solution.

		Color in acid	Color in base
Methyl orange	3.1 – 4.4		
Bromcresol green	3.8 – 5.4		
Methyl red	4.4 – 6.2		
Bromthymol blue	6.0 – 7.6		
Phenol red	6.8 – 8.4		
Phenolphthalein	8.3 – 10.0		

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Salt Hydrolysis

Salt	Parent Acid + Base	Dissociation	Result
KF	strong base (KOH) + weak acid (HF)	$KF(s) \rightarrow K^+(aq) + F^-(aq)$	Basic
NH_4Cl	weak base (NH_3) + strong acid (HCl)	$NH_4Cl(s) \rightarrow NH_4^+(aq) + Cl^-(aq)$	Acidic
$NaNO_3$	strong base (NaOH) + strong acid (HNO_3)	$NaNO_3(s) \rightarrow Na^+(aq) + NO_3^-(aq)$	Neutral

Rule of thumb: the ion from the weaker parent (acid or base) is the one that reacts with water.

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Buffered Solutions

- A buffer resists pH change when limited acid or base is added.
- Made of a weak acid/base and its conjugate — both ready to react.
- Buffer capacity: how much acid or base the buffer can absorb before pH shifts.
- Most effective when the conjugate pair's concentrations are equal or nearly equal.
- Blood must stay near pH 7.4 — the H_2CO_3/HCO_3^- buffer keeps it there.

Conjugate Pair	Buffer pH
HF / F^-	3.20
CH_3COOH / CH_3COO^-	4.76
H_2CO_3 / HCO_3^-	6.35
$H_2PO_4^- / HPO_4^{2-}$	7.21
NH_4^+ / NH_3	9.4

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Lesson Summary

Key Takeaways

1. Neutralization produces a salt and water: HCl + NaOH → NaCl + H_2O.

2. Titration finds unknown concentration by reacting a measured sample with a titrant to the equivalence point.

3. Indicators mark the visible end point; the best indicator's color change lines up with the equivalence point.

4. Salts of a weak acid or weak base hydrolyze, shifting pH; buffers use a conjugate pair to resist that shift.

Test Yourself

1. What is the molarity of a nitric acid solution if 43.33 mL of 0.1000M KOH is needed to neutralize 20.00 mL of the acid?

2. What is the concentration of a household ammonia solution if 49.90 mL of 0.5900M HCl neutralizes 25.00 mL of it?

3. Challenge: how many milliliters of 0.500M NaOH would neutralize 25.00 mL of 0.100M H_3PO_4?

4. Predict whether solutions of KF, NH_4Cl, and $NaNO_3$ are acidic, basic, or neutral, and explain why.

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